Mohamad Amin Mohamadi

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Research Interests

- Foundations of Machine Learning
- Statistical Machine Learning
- Theoretical Deep Learning

Education

- **University of British Columbia** — Vancouver, Canada
  Masters of Science in Computer Science; GPA: 93.5/100 — *Sep 2020 - Present*
  - **Supervisor:** Prof. Danica J. Sutherland
  - **Focus:** Working on theoretical deep learning with a focus on developing theoretical understanding of different aspects of neural networks through the lens of Neural Tangent Kernels and Tensor Programs.
  - **Relevant Courses:** Modern Statistical Learning Theory (A+), Optimization Theory (A+), Advanced Machine Learning (A+), Reinforcement Learning (A+)
- **Amirkabir University of Technology** — Tehran, Iran
  Bachelor of Science in Computer (Software) Engineering; GPA: 18.67/20 (Second in class of 140) — *Sep 2015 - Jun 2020*

Publications

- Making Look-Ahead Active Learning Strategies Feasible with Neural Tangent Kernel (arXiv/code)
  Mohamad Amin Mohamadi*, Wonhoe Bae*, Danica Sutherland
  *Neural Information Processing Systems (**NeurIPS**), 2022*
  (Recipient of the **NeurIPS 2022 Scholar Award**)
- A Fast, Well-Founded Approximation to the Empirical Neural Tangent Kernel (arXiv)
  Mohamad Amin Mohamadi, Danica Sutherland
  *Submitted to the International Conference on Learning Representations (**ICLR**), 2023*

Research Experience

- **Research Assistant Intern** — Joensuu, Finland
  University of Eastern Finland — *Jul - Aug 2019*
  Developed a novel algorithm (KM-ϵ) in pursuit of optimal categorical clustering of categorical data in linear time based on entropy purity measure. **Supervisor:** Prof. Pasi Franti
- **Research Assistant Intern** — Hong Kong, Hong Kong
  Hong Kong Baptist University — *Aug - Sep 2018*
  Developed a novel technique to locally interpret black-box machine learning models in the field of machine learning interpretability. **Supervisor:** Prof. Liang Lan

Work Experience

- **Oracle Labs** — Vancouver, Canada
  Research Scientist Intern (part-time) — *Nov 2021 - May 2022*
  - **AutoML for Time Series**: As a research intern in the AutoMLx team, successfully implemented, tested and deployed a forecasting model for general time series data based on the LightGBM model that improved the overall performance of the forecasting pipeline by more than 10% (based on internal forecasting benchmarks).
- **Cafebazaar** — Tehran, Iran
  Data Scientist — *May 2019 - Sep 2020*
  - **User Profiling**: Created a bunch of services to collect data from more than 40 million users, and organized them using *PySpark* and services for analyzing the basic behaviors of the users like funnels, show watching time duration and user segmentation for targeted promotions.
  - **Online Scalable Recommender System for Applications**: Re-implemented model trainer and hyper-parameter tuning using implicit-als algorithm as being a member of the team. Final output could handle over 500 requests per second.

Honors and Awards

- Recipient of the **NeurIPS 2022 Scholar Award**.
- Ranked **2nd among 140 undergraduate students** (GPA based) in Computer Engineering department, Amirkabir University of Technology, Tehran, Iran.
- Third place - Rescue Simulation challenge (SOSVR team) at **RoboCup 2016, Leipzig**
- Ranked in **top 0.2%** among all students in nationwide university entrance exam (Approximately 250000 applicants) in Math. and Eng., Iran, 2015.
- Awarded summer internship grant from **Prof. Fabian Theis (Technical University of Munich)**, 2019.

Invited Talks

- **A tale of ∞-wide neural networks** (slides)
  Math of Information, Learning and Data (MILD) - UBC, Sep 2022
- **A Fast, Well-Founded Approximation to the Empirical Neural Tangent Kernel** (slides)
  Privacy Preserving Machine Learning Group - UBC, Jul 2022
- **Neural Tangent Kernels** (slides)
  Machine Learning Reading Group (MLRG) - UBC, Mar 2022

Notable Academic Projects

- **Amortized Inference with Rejection Loops for Autonomous Vehicles**: Created a probabilistic generative model for traffic at road intersections using the CARLA simulation engine and PyProb. Performed amortized inference over the number of cars in the scene using inference compilation and amortized rejection loop-based importance sampling.
- **Variance-Reduced Methods for Neural Networks**: Novel implementation of various variance-reduced algorithms using PyTorch Optimizer's plug-in format and conducting experiments on their performance and convergence rate in comparison to traditional optimizers. The optimizers can be used for arbitrary complex PyTorch neural network architectures and are made open-source.
- **Muli-agent DRQN with Communication on PettingZoo Simple Tag Environment**: Successfully implement DRQN with a communication channel in a dynamical partially observable environment and qualitatively observe early-stage evidence of communication and planning among agents.
- **Unconstrained Optimization**: Implementation of algorithms like line search (steepest descent, newton, quasi-newton), and trust-region (dogleg, Cauchy point) methods, using *Numpy*.
- **CUDA-Shazam (Massively Parallel Song Detector)**: Applying Fourier transform to a part of a song, and comparing it using cosine similarity to find the complete song in parallel on *CUDA* to achieve ms-level response time.

Teaching experience

- **University Of British Columbia** Vancouver, Canada
  Teaching Assistant
  - **Masters of Data Science**: Preparing course material, holding office hours and assisting students in labs for different courses such as: Machine learning for Data Science, Statistical Inference (I) and (II), Advanced Probability and Statistics, Introduction to Probability and Statistics.
  - **Computer Science**: Holding office hours and assisting students in labs for different courses such as: Machine Learning (CPSC 340).
- **Amirkabir University of Technology** Tehran, Iran
  Teaching Assistant
  - **Computer (Software) Engineering**: Preparing course material, holding office hours and assisting students in labs for different courses such as: Artificial Intelligence, Algorithm Design, Discrete Mathematics, Probability and Statistics.

Skills Summary

- **Languages**: Python, Java, C/C++, SQL, Bash, R, JavaScript, Kotlin, LaTeX
- **Tools**: **Jax**, Docker, Git, PySpark, Numba, PyTorch, Numba, Mobile Development, Node.js